

SOLO + HIERO-SOLO-ACTION

Deploy, manage, and automate a private Hiero / Hedera test network

SOLO CLI

GITHUB ACTION

KUBERNETES

CI / CD



SPEAKER

ABOUT ME

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NOAH

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- **Public testnet:** shared, rate-limited, can't reset, needs internet
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| **Solo** makes it one command. **hierio-solo-action** makes it one CI step.

HEDERA VS. HIERO

HEDERA

A **public distributed-ledger network**.
Uses **Hashgraph** consensus (gossip-about-gossip + virtual voting), not a traditional blockchain. Native coin: **HBAR**.

HIERO

The **open-source codebase** that powers Hedera, contributed to the **Linux Foundation** for community governance.

Rule of thumb: **Hiero** = the software · **Hedera** = the public network that runs it.

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If Hedera's public network is grid electricity, **Solo** is a generator in your garage, and the **action** is a robot that switches it on automatically.

TWO TOOLS, ONE STORY

ITEM	REPO	WHAT IT IS	PURPOSE
01	solo	CLI tool	Spin up & manage a private Hiero network on your machine
02	hiero-solo-action	GitHub Action	Run that network automatically inside CI

01 SOLO · ENGINE



02 ACTION · AUTOMATION

// Docs: solo.hiero.org

A HIERO NETWORK = SEVERAL SERVICES



// Solo's whole job: install and wire these together for you.

WHAT SOLO IS

"An opinionated CLI tool to deploy and manage standalone test networks."

WHAT IT DOES

Orchestrates consensus nodes, mirror nodes, explorers, and relays **via Helm charts** on **Kind** clusters.

Install

```
npm install -g  
@hiero-  
ledger/solo
```

Requires

Node.js ≥ 22, Docker

SELF-MANAGING TOOLCHAIN

Solo **auto-installs kubectl, Helm, and Kind** at deploy time, so you only need Docker + Node.

THE DEPENDENCY STACK



RESOURCES

≥ 12 GB RAM, ≥ 6 CPU cores (16 GB / 8 cores for multi-node).

Docker Desktop caps usage independently of your machine, so raise the limits under **Settings → Resources**.

ONE-SHOT: A NETWORK IN ONE COMMAND

```
solo one-shot single deploy
```

Deploys the **full local stack**:

- Consensus node + mirror node
- Explorer + JSON-RPC relay
- Funded test accounts
- Port-forwards to localhost

Default endpoints (Solo 0.66+):

Explorer	localhost:38080
Consensus gRPC	localhost:35211
Mirror REST	localhost:38081
JSON-RPC relay	localhost:37546

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```
// Slow machine? add --no-parallel-deploy (the  
parallel stages can race)
```

WHAT GETS DEPLOYED

COMPONENT	DESCRIPTION
Consensus Node	Hiero consensus node for processing transactions.
Mirror Node	Stores and serves historical transaction data.
Explorer UI	Web interface for viewing accounts and transactions.
JSON RPC Relay	Ethereum-compatible JSON RPC interface.

ACCOUNT KEY TYPES

ED25519

Hedera default

Native SDK apps

ECDSA

EVM / MetaMask compatible

Ethereum tooling

The GitHub Action creates **both** account types, funded with test HBAR, and exposes them as workflow outputs.

THE SAME NETWORK, INSIDE CI

A **composite GitHub Action** you can drop into any workflow:

- No manual Java / Node / Kind / Helm / Solo setup
- One step → a live, throwaway network
- Hands your tests **funded account credentials**

Default: consensus node only. Enable mirror node and relay via inputs.

```
- name: Setup Hiero Solo
  uses: hiero-ledger/hiero-solo-action@v0.21.0
  id: solo

- name: Run tests
  env:
    ACCOUNT_ID: ${ steps.solo.outputs.accountId }
  run: npm test
```


WHAT THE ACTION DOES, IN ORDER

STEP 01 PREPARE

Install Java, Node, Kind, Helm · clean prior runs

STEP 02 INSTALL SOLO

npm-install the Solo CLI and create a Kind cluster

STEP 03 DEPLOY NETWORK

Deploy consensus node · port-forward to localhost

STEP 04 MIRROR NODE

Optional: if `installMirrorNode` or `installRelay`

STEP 05 RELAY

Optional: if `installRelay`

STEP 06 ACCOUNTS

Create & fund `ECDSA + ED25519` · publish as outputs

INPUTS, OUTPUTS & REAL-WORLD USE

KEY INPUTS (ALL OPTIONAL)

installMirrorNode	false
installRelay	false
dualMode (2 nodes)	false
hbarAmount	10,000,000

OUTPUTS

- accountId / publicKey / privateKey (ED25519)
- ecdsa* & ed25519* explicit variants
- deployment name

Use case: run integration tests against a fresh, isolated network on every push.

CLOSE-OUT

ONE COMMAND LOCALLY. ONE CI STEP IN THE CLOUD.

SOLO ONE-SHOT

-->

FULL LOCAL STACK

-->

ACTION = MINIMAL + FLAGS

Solo for local dev · **hierosolo-action** for automated testing in CI

[SOLO.HIERO.ORG](https://solo.hiero.org)

